

Postgres 16 Upgrade Plan

Goal

Upgrade all production Postgres clusters from 14 to 16 with zero data loss and under five minutes of write downtime per cluster. The plan uses logical replication to a new 16 cluster, a verified cutover, and a rehearsed rollback.

Preparation

Inventory extensions (pg_stat_statements, postgis, pgvector) and confirm 16 compatibility.

Freeze schema migrations for the cutover window. Verify database backups and run a full backup verification restore drill the week before.

Connection pooling via PgBouncer must be drained gracefully: reduce pool sizes, then pause; application retries are validated in staging with chaos testing.

Cutover Steps

1. Provision the 16 cluster with Terraform modules and apply the same VPC config and IAM policies.
2. Start logical replication and wait for lag under one second.
3. Pause writes at PgBouncer.
4. Confirm replication drain, promote the 16 cluster, flip the DNS record.
5. Resume writes; monitor Prometheus alerts and Grafana dashboards for regression.

Rollback

Reverse replication runs from the new cluster back to 14 until the soak window closes, so rollback is a second cutover. Postgres vacuuming baselines and autovacuum settings are re-checked after promote; replication slots are dropped only after seven clean days.

References

Postgres 16 release notes